

National Coverage Determination (NCD)

Hyperbaric Oxygen Therapy

20.29

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Tracking Information

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20.29

Manual Section Title

Hyperbaric Oxygen Therapy

Version Number

2

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04/01/2003

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06/19/2006

Implementation Date

04/01/2003

Description Information

Benefit Category

Incident to a physician's professional Service

Outpatient Hospital Services Incident to a Physician's Service

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

For purposes of coverage under Medicare, hyperbaric oxygen (HBO) therapy is a modality in which the entire body is exposed to oxygen under increased atmospheric pressure.

Indications and Limitations of Coverage

A - Covered Conditions

Program reimbursement for HBO therapy will be limited to that which is administered in a chamber (including the one man unit) and is limited to the following conditions:

1. Acute carbon monoxide intoxication, (ICD-9-CM diagnosis 986).
2. Decompression illness, (ICD-9-CM diagnosis 993.2, 993.3).
3. Gas embolism, (ICD-9-CM diagnosis 958.0, 999.1).
4. Gas gangrene, (ICD-9-CM diagnosis 0400).
5. Acute traumatic peripheral ischemia. HBO therapy is a valuable adjunctive treatment to be used in combination with accepted standard therapeutic measures when loss of function, limb, or life is threatened. (ICD-9-CM diagnosis 902.53, 903.01, 903.1, 904.0, 904.41.)
6. Crush injuries and suturing of severed limbs. As in the previous conditions, HBO therapy would be an adjunctive treatment when loss of function, limb, or life is threatened. (ICD-9-CM diagnosis 927.00-927.03, 927.09-927.11, 927.20-927.21, 927.8-927.9, 928.00-928.01, 928.10-928.11, 928.20-928.21, 928.3, 928.8-928.9, 929.0, 929.9, 996.90-996.99.)
7. Progressive necrotizing infections (necrotizing fasciitis), (ICD-9-CM diagnosis 728.86).
8. Acute peripheral arterial insufficiency, (ICD-9-CM diagnosis 444.21, 444.22, 81).
9. Preparation and preservation of compromised skin grafts (not for primary management of wounds), (ICD-9CM diagnosis 996.52; excludes artificial skin graft).
10. Chronic refractory osteomyelitis, unresponsive to conventional medical and surgical management, (ICD-9-CM diagnosis 730.10-730.19).
11. Osteoradionecrosis as an adjunct to conventional treatment, (ICD-9-CM diagnosis 526.89).
12. Soft tissue radionecrosis as an adjunct to conventional treatment, (ICD-9-CM diagnosis 990).
13. Cyanide poisoning, (ICD-9-CM diagnosis 987.7, 989.0).
14. Actinomycosis, only as an adjunct to conventional therapy when the disease process is refractory to antibiotics and surgical treatment, (ICD-9-CM diagnosis 039.0-039.4, 039.8, 039.9).
15. Diabetic wounds of the lower extremities in patients who meet the following three criteria:
 - a. Patient has type I or type II diabetes and has a lower extremity wound that is due to diabetes;

- b. Patient has a wound classified as Wagner grade III or higher; and
- c. Patient has failed an adequate course of standard wound therapy.

The use of HBO therapy is covered as adjunctive therapy only after there are no measurable signs of healing for at least 30 –days of treatment with standard wound therapy and must be used in addition to standard wound care. Standard wound care in patients with diabetic wounds includes: assessment of a patient's vascular status and correction of any vascular problems in the affected limb if possible, optimization of nutritional status, optimization of glucose control, debridement by any means to remove devitalized tissue, maintenance of a clean, moist bed of granulation tissue with appropriate moist dressings, appropriate off-loading, and necessary treatment to resolve any infection that might be present. Failure to respond to standard wound care occurs when there are no measurable signs of healing for at least 30 consecutive days. Wounds must be evaluated at least every 30 days during administration of HBO therapy. Continued treatment with HBO therapy is not covered if measurable signs of healing have not been demonstrated within any 30-day period of treatment.

B - Noncovered Conditions

All other indications not specified under §35-10(A) are not covered under the Medicare program. No program payment may be made for any conditions other than those listed in §35-10 (A).

No program payment may be made for HBO in the treatment of the following conditions:

1. Cutaneous, decubitus, and stasis ulcers.
2. Chronic peripheral vascular insufficiency.
3. Anaerobic septicemia and infection other than clostridial.
4. Skin burns (thermal).
5. Senility.
6. Myocardial infarction.
7. Cardiogenic shock.
8. Sickle cell anemia.
9. Acute thermal and chemical pulmonary damage, i.e., smoke inhalation with pulmonary insufficiency.
10. Acute or chronic cerebral vascular insufficiency.
11. Hepatic necrosis.
12. Aerobic septicemia.
13. Nonvascular causes of chronic brain syndrome (Pick's disease, Alzheimer's disease, Korsakoff's disease).
14. Tetanus.
15. Systemic aerobic infection.

16. Organ transplantation.
17. Organ storage.
18. Pulmonary emphysema.
19. Exceptional blood loss anemia.
20. Multiple Sclerosis.
21. Arthritic Diseases.
22. Acute cerebral edema.

C - Topical Application of Oxygen

This method of administering oxygen does not meet the definition of HBO therapy as stated above. Also, its clinical efficacy has not been established. Therefore, no Medicare reimbursement may be made for the topical application of oxygen.

Cross Reference

See section 270.5 of the NCD Manual.

Claims Processing Instructions

[TN 187 \(Medicare Claims Processing Manual\)](#) 

[TN 3921 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

164

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R164CIM.pdf> 

Revision History

12/27/2002 - Expanded coverage for treatment of diabetic wounds of the lower extremities in patients that meet three criteria. Effective date 04/01/2003. ([TN 164](#) ) (CR 2388)

10/19/2000 - Manualized program memorandum AB-00-15 (dated 4/1/2000) and clarified that "preparation and preservation of compromised skin graft" in section 35-10A.9 is not for primary management of wounds. Effective date NA. ([TN 129](#) ) (CR 1138)

04/01/1999 - Clarified covered conditions and physician supervision requirement. Effective date 05/01/1999. (TN 112)

07/01/1997 - Clarified coverage limited to conditions listed under §35-10.A. Effective date 08/11/1997. (TN 102)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Hyperbaric Oxygen Therapy for Hypoxic Wounds and Diabetic Wounds of the Lower Extremities \(CAG-00060N\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Hyperbaric Oxygen Therapy	4	04/03/2017 - N/A	View
Hyperbaric Oxygen Therapy	3	06/19/2006 - 04/03/2017	View
Hyperbaric Oxygen Therapy	2	04/01/2003 - 06/19/2006	You are here
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